Imputation of bulk density, soil pH, and texture measurements

Supporting document `IndigoCarbon_US-1_2023_0054`

Indigo Ag

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Abstract

This document summarizes how Indigo imputes bulk density, soil pH, and texture at sample points. Because of the costs and lower variability of these soil attributes, they are measured at a random subset of points (and, in the case of pH and texture, using composite samples). Geostatistical techniques are used to estimate those measurements at sample points. This document shows how this imputation satisfies the requirements for imputing soil measurements defined in the supporting document *Addressing incomplete soil sample data in Soil Enrichment Protocol projects* (`IndigoCarbon_US-1_2023_0062`, 2023) that was reviewed by an independent expert.

Changes in this supporting document since the last verification

New soil samples taken since the last verification have been incorporated into the analysis: the imputation models were re-fitted, and the assessment of the models has been refreshed accordingly. Other than re-fitting the models to a new training dataset, the models have not been changed.

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1 Introduction

As described in Table 6.2 of the SEP, the soil measurements must be taken at randomly selected locations (points) according to a sample design with the aim of collecting a representative sample of soils. The biogeochemical models estimate emissions reduction at each sampled point.

In accordance with the guidance document *Addressing incomplete soil sample data in Soil Enrichment Protocol projects* (`IndigoCarbon_US-1_2023_0062`, 2023) (hereafter, "guidance document"), bulk density, pH, and texture may be measured at a random subset of points where %SOC is measured because they are expensive to take, lead to more injuries to soil samplers (in

the case of bulk density), and/or vary less within fields than %SOC does (Franzluebbers, 2010). At points where these measurements were missing—either deliberately so (to reduce costs) or due to unforeseen challenges (e.g., sample lost in shipment; part of a field was flooded)—they can be imputed using models, subject to the requirements in Section 5.3 of the guidance document.

Table 1, which replicates a table in the Sample Design section of the Monitoring Plan, shows how samples are assigned to sample points. The parts in bold font are where imputation is needed to make predictions because a sample was not collected.

Table 1: How samples were randomly assigned to points. Changes in the way bulk density, pH, and texture samples were assigned to points are highlighted in bold font.

Timeframe	Sample design	**How samples are randomly assigned to points**		
		SOC	Bulk density	pH/Texture
Oct. 1, 2019 to Mar. 16, 2020	POISSWR (national point map)	All points	1 in 5 points	Not collected
Mar. 16, 2020 to Aug. 31, 2020	POISSWR (national point map)	All points	**1 in 3 points**	**Composite sample at bulk density points**
Sep 13, 2020 onward	Two-stage design	All points	**1 in 3 points**	

This document describes Indigo's approach to imputation for these soil properties that are sampled less densely than soil organic carbon concentration is. Indigo's approach to handling missing soil organic carbon measurements is addressed in a separate supporting document (`IndigoCarbon_US-1_2023_0063`, 2023).

To develop our imputation models for bulk density, pH, and texture, we first followed the criteria laid out in section 5.3 of the guidance document for handling missing samples. Among the developed imputation models that meet these criteria, we select the highest performing model for each soil property as determined through the root mean squared error (RMSE) of cross-validation experiments using available soil measurements and other potential predictors of these properties.

For predicting bulk density, we fit a regression kriging model that uses all available bulk density measurements. For each texture property and for soil pH, we train linear regression models using (a) point-level soil measurements taken outside of the Carbon by Indigo program and (b) digital soil survey data.

We impute these soil properties at every sample location with a soil organic carbon measurement but missing any of the other soil properties. Predictions of each soil property are made using a random draw from the model's predictive distribution at that sample location. By using a draw from the predictive distribution (rather than using the mean prediction at each point), we reflect our uncertainty about imputed values and propagate that uncertainty through the biogeochemical model. Those draws from predictive distributions are also required by the guidance document, and they enable us to introduce variability into our predictions so that they match that of measured properties (another requirement of the guidance document).

The remainder of this supporting document is structured as follows. In Section 2, we present an assessment of our imputation models for each of the soil properties, bulk density, soil pH, clay percent, and sand percent (silt percent is calculated as 100 less the clay percent less than sand percent) in accordance with Section 3.5 of the guidance document. In Section 3, we discuss our imputation of bulk density, beginning with the results of the imputation model in Section 3.1 followed by a summary of that imputation model in Section 3.1.1.

2 Assessment of imputation models

The guidance document allows us to impute bulk density, soil texture, and soil pH wherever they are missing. A summary of the provided guidance for these imputation models is presented below:

- Point-level imputations may be done with a single draw from a predictive distribution to maintain similar within-field variability as would be observed with no missing data. Predictive distributions may be constructed analytically, or with a resampling or simulation method, so long as the method has a theoretical basis.
- External datasets may be used to train imputation models, including national soil surveys, remote sensing data, and soil samples collected outside of the sample project
- For each imputation model, report summaries of the imputed values must be provided including
  - Summaries of the number of missing measurements
  - Variability of imputed values
  - Associations between soil attributes
  - Performance of the imputation model

We discuss this guidance and criteria in the context of each imputation model below.

3 Imputation of bulk density

We begin in Section 3.1 by describing the performance of our chosen imputation model for bulk density. The choice of this model among many alternatives is described later in Section 3.2 and 3.3.

3.1 Summary of the performance of the bulk density imputation model

3.1.1 Bulk density imputation model

For bulk density imputation, we use a regression kriging model.

Regression step The first step of the model is a linear regression

$$\texttt{bulk_density} = 1.535 + -0.085 \times \texttt{soil_carbon_concentration} \tag{1}$$

The negative correlation between bulk density and the concentration of SOC, captured by the negative coefficient -0.085 in Equation (1), is known in literature (see, e.g., Keller and Håkansson (2010, Figure 4, Section 3.2)). The linear regression model (1) has coefficient of determination $R^2 = 0.145$; note that this R^2 is just from the first step of the model (linear regression), and the value of R^2 is higher when one also uses the second step of the model (kriging), described next.

Table 2: Summary of the kriging step of the bulk density imputation model. The final model had ID 1653437264, which is a hash of the data used to fit the model and the hyperparameters. The model was fit on June 15, 2024.

maximum lag (meters)	6000
variogram model	spherical
number of lags (i.e., bins)	60
nugget $(g/cm^3)^2$	0.008
range of the variogram (meters)	3215.451
sill $(g/cm^3)^2$	0.006

Kriging step The second step of the bulk density imputation model is kriging of the residuals of the linear regression model (1). The crux of this kriging step is the variogram, which captures the tendency for these residuals to be similar at nearby locations (i.e., the spatial autocorrelation). The parameters of this kriging model are shown in 2. Collectively, the linear regression and kriging model have coefficient of determination $R^2 = 0.796$. (Note that the model explains 79.6% of the variance, and draws from the predictive distribution (described in Sec. 3.1.2) ensure that uncertainty is not under-estimated.)

The variogram of the kriging model has a "maximum lag" parameter set to 6000 meters (see Table 2). For points with no bulk density measurement within 6000 meters, the kriging model uses the sill as the estimate of the variance, and a warning is given by the SciKit-Gstat package. This situation occurred for 118 points, in fields that are small and far from other fields.

3.1.2 Predictive distribution

As required by the guidance document `IndigoCarbon_US-1_2023_0062` (2023), draws from a predictive distribution must be used to capture the uncertainty we have about imputed values. For each imputed bulk density sample point, the predictive distribution is centered at the mean bulk density prediction calculated from the fit regression kriging model, with standard deviation equal to the square root of the kriging error variance at that location. (For documentation of the kriging variance, see *SciKit-GStat* (2021), a Python package for kriging used in our model.) In the event a prediction was less than or equal to zero after this draw from the predictive distribution, that prediction was replaced by the minimum positive prediction in the dataset (see the row labeled "minimum" and column labeled "BD is imputed" in Table 6 in Section 3.1.5). This replacement occurred for 1 point.

Table 3 summarizes the predictive distribution. The average kriging error used for our predictions was 0.101 g/cm^3, or 7.1% of the mean bulk density prediction of 1.412 g/cm^3. Taking draws from this predictive distribution increases the overall variability of our predictions over mean predictions: overall standard deviation increases by 24% and average within-field standard deviation increases by 5%.

Table 3: **Summary of the predictive distribution** of the bulk density imputation. All units are g/cm^3. Abbreviation: SD = standard deviation.

Quantity	Summary statistic	Value
mean prediction	average across points	1.41
	SD across points	0.14
	average of within-field SDs	0.14
SD of the predictive distribution	averaged across points	0.10
	SD across points	0.01
single draw from the predictive distribution	average across points	1.41
	SD across points	0.17
	average of within-field SDs	0.17

3.1.3 Correlation with other soil properties

For bulk density, we capture the correlation between bulk density and soil carbon concentration through the regression component of our regression kriging algorithm. As a result, the Spearman correlations (Table 4) are fairly similar, as required by the guidance document `IndigoCarbon_US-1_2023_0062` (2023).

Table 4: The **Spearman correlation** between bulk density and soil organic carbon concentration is similar for points where bulk density is measured and for points where bulk density is imputed.

	Spearman correlation of BD and %SOC
Points with measured bulk density	-0.399
Points with imputed bulk density	-0.368

3.1.4 Number of points and fields at which bulk density is imputed

For bulk density imputation, we use measurements taken in the Carbon by Indigo program. No measurements in other programs are used.

The number of such measurements is listed in Table 5. Note that the number of fields with soil samples is larger than the number of fields on which credits are being issued, because some fields dropped out of the program after they were soil sampled. Those fields are still useful in the kriging model because they help to better identify the statistical relationships between soil organic carbon concentration and bulk density (in the regression step of the imputation model) and in the spatial

autocorrelation of the residuals of that linear regression (in the kriging step of the imputation model).

Table 5: Summary of the number of bulk density measurements that were imputed.

Sample points with imputed bulk density	49110 of 72127 (68.1%)
Fields with ≥ 1 imputed bulk density	5690 of 6120 (93.0%)

3.1.5 Variability of imputed and measured bulk density values

Table 6 summarizes the variability of the measured and imputed values of bulk density. Here, the imputed values are the (single) draws from the predictive distribution, the same values provided to the biogeochemical model. Notice that the distributions are similar in terms of their quartiles, average, and standard deviation. The typical variability within fields ("average of the within-field SDs") are also similar. This table shows that the imputations are accurately capturing the variability in the ground-truth measurements, as required by the guidance document `IndigoCarbon_US-1_2023_0062` (2023).

Table 6: Summary of the variability of bulk density measurements and imputed values. All units are g/cm^3. Abbreviation: SD = standard deviation; BD = bulk density.

Points where bulk density is...	imputed	measured
minimum	0.38	0.60
25th percentile	1.31	1.32
50th percentile	1.42	1.42
average	1.41	1.41
75th percentile	1.53	1.52
maximum	2.14	1.80
SD	0.18	0.17
average of the within-field SDs	0.11	0.10

3.1.6 Bias and accuracy in cross validation

To assess the ability of the imputation model to predict accurately and precisely out-of-sample, we conduct k-fold cross validation with $k = 10$ splits (as explained in Section 3.1.6). The accuracy and bias on held-out data is shown in Table 7. Importantly, the bias of the predictions is negligible.

3.2 Methodology for selecting a model to predict bulk density

We use available soil measurements from the Carbon by Indigo program within a cross validation framework to test a suite of bulk density prediction models. These models are evaluated for bias and performance before our final model is selected. For use in the Carbon by Indigo program, this final model is trained on all available bulk density samples taken in the Carbon by Indigo

Table 7: Summary of the performance of the bulk density imputation model on held-out data in cross validation. Shown is the average of the metric, plus or minus the standard deviation, across the $k = 10$ folds. Abbreviation: RMSE = root mean squared error.

RMSE	0.107 +/- 0.002 g/cm^3
Bias	0.000 +/- 0.002 g/cm^3

program and is used to make bulk density predictions at all sample locations without bulk density measurements.[1] In this section we walk through our analysis datasets, cross validation framework, considered models, and considered predictors. Results of this model selection process are presented in Section 3.3.

3.2.1 Analysis dataset

We use a subset of all soil sample locations for model selection. Table 8 provides a summary of this subset and our justification for each limitation

Table 8: Exclusion criteria for the cross-validation of the bulk density imputation model

Exclusion criteria	Justification	Points
All points	N/A	72127
All points with a bulk density measurement	Model and cross validation can only be fit and evaluated with measured values	23017
All points in fields with ≥ 2 bulk density measurements	See Cross validation framework (Section 3.2.2) for explanation of the need for 2+ measurements per field	22102

3.2.2 Cross validation framework

Our aim with predicting bulk density is to use bulk density measurements within each field (and nearby fields) to predict at SOC sample points in the field. We sought a cross validation framework to simulate and test this approach of predicting using nearby points.

For cross validation, we use a modification on shuffle-split cross validation in which we withhold one sample point from every field to use as the test dataset. All other points are used in the train dataset. We call this splitting method, GroupShuffleSplit. This splitting approach simulates a world in which we have at least one bulk density sample point in each field, which is a property of our actual sampling scheme. Figure 1 visually depicts this hold-out scheme.

We use two metrics to evaluate and compare the performance of the models considered for predicting bulk density. Table 9 describes the scoring in more detail.

[1] This practice—of doing (1) cross-validation for model selection and assessment and (2) fitting the model on *all available training data* (rather than a randomsample of, say, 80% of the training data)—is common outside of "Big Data" contexts: because there are preciously few samples, we wish to all available measurements for training the model that serves predictions in production. Because the cross-validation framework uses almost as many training data points as this production model, the estimated precision and accuracy on held-out data (see Tables 9 and 7) give reliable estimates of the out-of-sample performance of the production model.

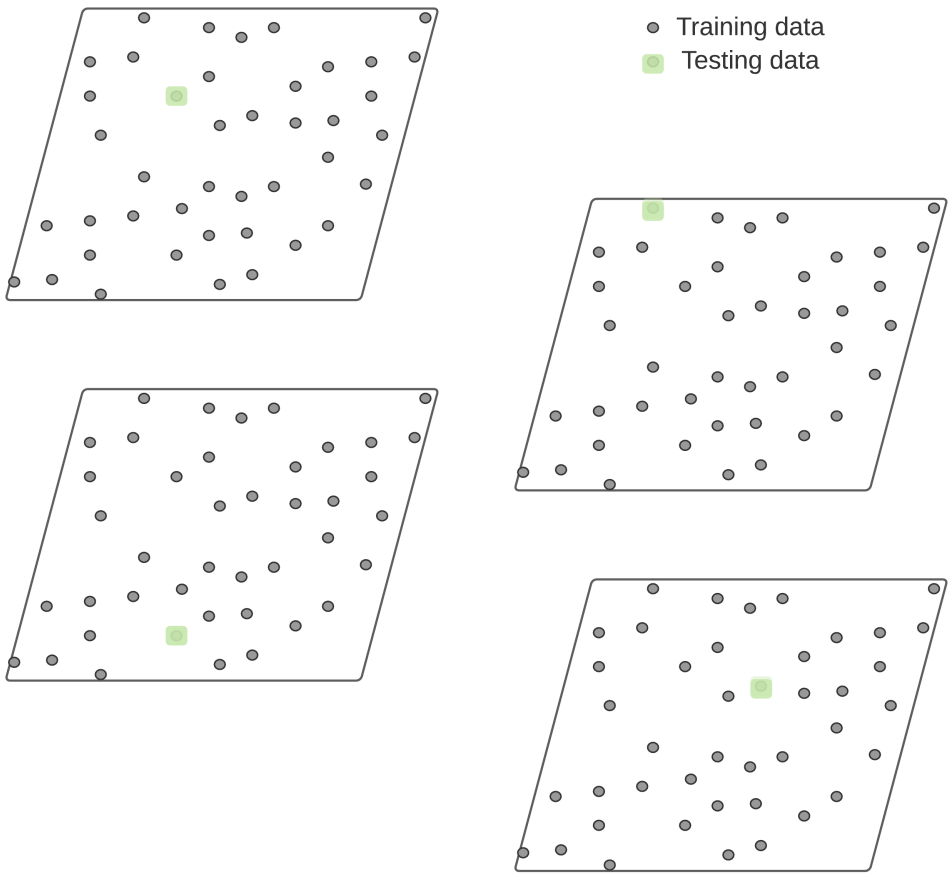


Figure 1: **Illustration of the hold-out scheme for cross validation of the bulk density imputation model.** Because we impute in fields where we have bulk density measurements taken elsewhere in the field, we simulate such prediction tasks by hold out one random point per field.

Table 9: Scoring metrics

Metric	Definition
Bias	The average of the errors. Used to determine systematic difference between the prediction and the true value.
RMSE	Root Mean Squared Error (RMSE) is the square root of the average squared error. RMSE gives more weight to larger errors (compared to other common metrics like mean absolute error)

3.2.3 Models considered

We compare all models considered for predicting bulk density to each other and to the baseline scenarios of using the simple global average of bulk density and using the simple field-level averages of bulk density. The models we considered for interpolation ranged from distance-weighting approaches to linear regression analysis, machine learning algorithms, and kriging approaches. Many of these models also use hyperparameters, or configurations of the model whose values cannot be estimated from data. These hyperparameters can further be used to optimize model performance. Table 10 summarizes the models we considered as well as their hyperparameters.

3.2.4 Predictors considered

In our evaluation of models to predict bulk density, we considered several categories of predictors. We may also consider new predictors in future iterations of our bulk density predictions. Table 11 below describes these considered predictors.

3.3 Results of model selection for predicting bulk density

3.3.1 Selected approach

We find that regression kriging leads to the best estimations of bulk density. In our cross validation experiments, regression kriging with a simple linear regression of bulk density on soil carbon concentration produced the lowest root mean squared error of predicted values. Table 12 summarizes the performance of our considered models.

While we find that including gSSURGO predictors improves predictions in linear regression and machine learning algorithms, we do not observe this same improvement with regression kriging and selected a regression kriging approach whose regression is only a linear regression on measured SOC.

3.3.2 Validating assumptions of our variogram

The experimental variogram used in regression kriging should only be computed and used out to a spatial distance that is logical for the spatial property being interpolated (Oliver and Webster, 2015, p. 37). We select a maximum lag over which to fit our variogram so that the fitted model is accurate over the distance over which we want to krige. We test distances between 2000–7000 meters which is approximately the diameter of 4 to 14 adjacent 80-acre fields. The lag determined in our variogram computing step is described in Table 2 above. What happens over distances larger than that across the maximum lag is of little consequence for the predictions (i.e., a point 12 80-acre fields away should have little correlation with the point we are trying to predict).

When constructing our variogram for the regression kriging model we considered many of the major variogram assumptions:

1. We find no **skewness** in the data, so no transformation was needed to ensure normality and adjust for outliers.

2. We do not observe major trends in our data across space, nor do we observe differences between the x (longitude) direction and y (latitude) direction, so we do not adjust for **anisotropy** and believe our assumption about **weak stationarity** holds.

Table 10: Models considered for the bulk density imputation.

Model	Hyperparameters	About / why considered
Global mean		Simplest interpolation approach serving as baseline
Field-level mean		Simple interpolation approach serving as baseline
gSSURGO		values of bulk density from *gSSURGO* (2020) (a gridded version of SSURGO) at each sample location serving as baseline
K-nearest neighbors	• Number of neighbors • Weighting approach • Power parameter for weighting	Intuitive and efficient approach using proximal measurements data
Linear regression		Straightforward, easy to interpret global estimator with interpretable estimates of error
Decision Tree regression	• Maximum tree depth • Minimum samples at a leaf • Maximum number of leaf nodes	Simple machine learning algorithm
AdaBoost	• Estimator • Loss function for updating weights • Learning rate contribution of each estimator	Complex meta-estimator of a machine learning algorithm that fits multiple copies of the same estimator, tuning in each repetition
Ordinary kriging	• Variogram model • Kriging specifications	Common geospatial technique that imputes values using a statistical model that describes covariability over space
Regression kriging	• Ordinary kriging hyperparameters • Regression model used prior to kriging of errors	This procedure was suggested by Ahmed and De Marsily (1987) and Odeh, McBratney, and Chittleborough (1995) where the drift and residuals can be estimated separately and then summed

Table 11: Predictors considered for the bulk density imputation model

Predictor category	Predictor	About / why considered
Location	Latitude and longitude	Used for geographic trends and models that use proximal points for predictions
	Location x and y coordinates in Albers projection	Same reasons as above, but locations are transformed to an equal-area projection (in meters)
Soil measurements	Soil organic carbon	Measured SOC at each point, correlated with bulk density and available at every bulk density measured and non-measured location
gSSURGO	Bulk density	Bulk density from gSSURGO survey
	Soil organic matter	SOM from gSSURGO survey
	Texture	Sand, Clay, and Silt percentages from gSSURGO survey
	Soil pH	Soil pH from gSSURGO survey

When a point we need to interpolate is further than the maximum lag from any other bulk density measurement, our algorithm uses the average error of the upstream linear model instead of the kriging estimate of the nearby errors that would otherwise be used in regression kriging.

This scenario would only happen on fields where there are no bulk density measurements (and no bulk density measurements in nearby fields). The number of such cases is 118, as mentioned in Section 3.1 above. The kriging error at these sample location is undefined. As a conservative estimate, we set the predictive distribution in these scenarios to the sum of the nugget and the sill of the variogram used in our kriging step. This sum reflects the co-variability of points beyond the distance at which the points are expected to be spatially dependent (Oliver and Webster, 2015, p. 31).

4 Imputation of soil pH and texture

4.1 Assessment of soil pH imputation model

For pH imputation, we use a linear regression model described in more detail below. This model is trained on an external dataset of point-level soil pH measurements, synthetically constructed field-level composite soil pH measurements, and point-level gSSURGO values.

4.1.1 Imputation model

For soil pH, we use a linear regression model with two predictors:

Table 12: Results of cross validation experiments for predicting bulk density. Our chosen model, highlighted in green, achieves the lowest RMSE. Abbreviations: RMSE = root mean squared error; SOC = soil organic carbon concentration. Units of the bias and RMSE columns are both g/cm^3. The approaches marked with * above used *nested* cross-validation: the model itself does its own cross-validation on the training set to choose optimal hyperparameters. Thus, the optimal hyperparameters vary from one fold to another (of the outer k-fold cross validation).

Interpolation approach	Bias	RMSE
AdaBoost (SOC, location, & gSSURGO soil properties)*	-0.0005	0.0885
AdaBoost (SOC, location)*	0.0004	0.0908
K-Nearest Neighbors*	-0.001	0.0921
Decision Tree Regressor (SOC, location)*	0.0021	0.0983
Decision Tree Regressor (SOC, location, & gSSURGO soil properties)*	0.0026	0.099
Field-level mean	-0.0003	0.1025
Linear regression (SOC, location, & gSSURGO soil properties)	0.0043	0.106
Linear regression (SOC, location)	0.0048	0.1164
Linear regression (SOC)	-0.0059	0.1183
Global average	-0.0082	0.1416
Regression kriging (SOC)*	0.0008	0.0857
Regression kriging (SOC, location, all gSSURGO soil variables)*	0.0003	0.0873
Ordinary kriging*	-0.0004	0.0919
gSSURGO	0.121	0.178

1. the soil pH according to gSSURGO (denoted `soil_pH_SSURGO`) and
2. the composite measurement of soil pH taken by Indigo (denoted `soil_pH_composite_sample`).

The fitted regression model is:

$$\texttt{pH} = 0.185 + 0.801 \times \texttt{soil_pH_composite_sample} + 0.197 \times \texttt{soil_pH_SSURGO} \tag{2}$$

Notice in Equation (2) that more weight is assigned to the ground-truth measurement in the field than to the prediction from SSURGO, indicating that even a single composite sample of pH that represents the field adds a lot of predictive value over using SSURGO alone. The coefficient of determination of the model (2) is $R^2 = 0.403$. The model had ID `3440608739` (this is a hash of the data used to fit the model and of the hyperparameters), and it was fit on June 15, 2024.

4.1.2 Predictive distribution

For each imputed sample point, the predictive distribution is normally distributed with mean equal to the soil pH prediction from the linear regression model and with standard deviation equal to the square root of the calculated prediction variance for the predicted point.

Table 13 summarizes the predictive distribution. The standard deviation of the predictive distribution is typically 0.62. To put that uncertainty of each imputation in perspective, that is 9% of the average imputed value of pH, 6.71.

Table 13: **Summary of the predictive distribution** of the soil pH imputation. Abbreviation: SD = standard deviation.

Quantity	Summary statistic	Value
mean prediction	average across points	6.70
	SD across points	0.66
	average of within-field SDs	0.24
SD of the predictive distribution	averaged across points	0.62
	SD across points	0.00
single draw from the predictive distribution	average across points	6.71
	SD across points	0.80
	average of within-field SDs	0.60

In rare cases, this normal distribution produces pH values that are outside the typical range found in cropland, due to the tails on normal distributions. The USDA NRCS reports in *Soil Quality Indicators: pH* (1998) that soil pH below 5.6 is considered low for most crops, and pH above 8.0 is considered high for most crops. In the training dataset (discussed in Sec. 4.1.4 below), 14% of pH measurements are below 5.6, and 2.6% of pH measurements are greater than 8. Thus, to make a range of agronomically plausible pH values, we chose $[5, 8.5]$ for North American cropland; in the training dataset, these values are the 1.7% percentile and 99.6% percentile. To avoid providing implausible values of pH to the biogeochemical model, we ensure that our imputations fall within this agronomically plausible range by conditioning the predictive distribution on falling in the range $[5, 8.5]$ (i.e., we reject draws from the predictive distribution that fall outside that range).

4.1.3 Correlation with other imputed soil properties

For soil pH imputation, we predict pH using the composite measurement of pH and gSSRUGO's prediction of pH at the point. The model does not explicitly capture correlation with other soil properties.

4.1.4 Datasets used

For soil pH imputation, we train our model with a static dataset of point-level soil pH measurements. This dataset was collected by Indigo in 2018 for business purposes outside of the Carbon by Indigo program called *Indigo Research Partners*, which did 1-acre grid sampling of soil pH in commercial fields undergoing field trials. See section 4.3 for an overview of the overlap in state coverage and soil property values between Indigo Research Partners and the Carbon by Indigo program.

With these point-level soil pH measurements from Indigo Research Partners, we construct field-level composite measurements from a random subsample of measurements in each field. This composite measurement is used as a predictor variable; it serves as a proxy for the composite samples in the Carbon by Indigo program.

In addition to the point and field-level soil pH measurements, we use point-level gSSURGO estimates for soil pH. These estimates are the weighted average gSSURGO major component values down to 30 centimeter depths. gSSURGO values were imputed from layers above or below layers with missing gSSURGO values.

gSSURGO values were missing for a relatively small number of points in both the training dataset in Indigo Research Partners and in the prediction dataset in the Indigo project (Table 14). These points were dropped from the analysis: points in Indigo Research Partners that were missing gSSURGO were not used to fit the imputation models, and points in the project that were missing gSSURGO were not sent to the biogeochemical model (but were captured in downstream analyses of bias in `IndigoCarbon_US-1_2023_0062`, 2023). An example situation that leads to gSSURGO values being missing is when the map unit is a "borrow pit": the land may have been scraped away many years ago (e.g., to build a highway overpass), and it was recently replanted. Soil databases like gSSURGO cannot reliably predict the soil profiles in land with such disruptions.

Table 14: The number of points in the training dataset and prediction dataset where gSSURGO was missing.

IRP (training dataset)	20 of 49592 (0.04%)
Carbon by Indigo (prediction dataset)	73 of 72475 (0.1%)

If a field in the Carbon by Indigo program does not have a recorded pH or texture composite sample, then we guess the composite measurement value that we would have obtained had we pulled a composite sample from the field by averaging gSSURGO values at all locations where SOC concentration was measured on the field.

4.1.5 Number of points and fields at which soil pH is imputed

Table 15 lists the number of fields in which imputations of soil pH were made. Note that this number of fields is larger than the number of fields in the issuance because more fields were soil sampled than ended up in the issuance (due to attrition), as mentioned in Section 3.1.4. Because

those sampled fields are still useful for the kriging step in the bulk density model in Section 3, we impute pH in those fields, as well, even though this model does not improve as the size of the Carbon by Indigo program grows. Instead, the pH imputation model improves with the size of its training dataset, which is Indigo Research Partners and gSSURGO predictions on those fields (as described in Section 4.1.4).

Table 15: Summary of the number of soil pH measurements that were imputed. Abbreviation: SD = standard deviation.

Fields	6161
Carbon points per field (mean ± SD)	11.8 ± 8.1
Fields with composite samples	4630 (75.1%)

4.1.6 Variability of imputed and measured soil pH values

Table 16 summarizes the variability of the measured and imputed values of soil pH in the training dataset (Indigo Research Partners) and in the imputations made in the Carbon by Indigo program. As required by the guidance document `IndigoCarbon_US-1_2023_0062` (2023), the imputed values are the (single) draws from the predictive distribution, so that uncertainty about imputed values is propagated to the biogeochemical model. The distributions are fairly similar. (They are not as similar as they were for bulk density (Table 6) because here the model must generalize from one dataset (Indigo Research Partners) to another (Carbon by Indigo) and has only composite-level measurements, whereas the bulk density model can exploit measurements at individual points in the Carbon by Indigo program.) Importantly, the typical variability within fields ("average of the within-field SDs") are similar, and the imputed values are typically more variable within fields, indicating that uncertainty is being estimated conservatively.

Table 16: Summary of the variability of soil pH measurements and imputed values. Abbreviation: SD = standard deviation.

Soil pH measured in...	Carbon by Indigo	Indigo Partners
minimum	5.000	1.000
25th percentile	6.110	5.800
50th percentile	6.691	6.400
average	6.706	6.410
75th percentile	7.293	6.900
maximum	8.500	13.000
SD	0.801	0.796
average of the within-field SDs	0.601	0.540

4.1.7 Bias and accuracy in cross validation

To assess the ability of the imputation model to generalize outside the Indigo Research Partners training dataset accurately and precisely, we conduct k-fold cross validation with $k = 10$ splits (as

explained in Section 4.3.2). The accuracy and bias on held-out data is shown in Table 17. The bias of the predictions is negligible.

Table 17: Summary of the performance of the soil pH imputation model on held-out data in cross validation. Shown is the average of the metric, plus or minus the standard deviation, across the $k = 10$ folds. Abbreviation: RMSE = root mean squared error.

RMSE	0.615 +/- 0.003
Bias	0.001 +/- 0.004

4.2 Assessment of soil texture imputation model

4.2.1 Imputation model

For imputation of soil texture, we fit two separate linear regressions with logit-transformed outcomes, one for sand percent and one for clay percent, described in more detail in Section 3. Imputed silt values are estimated as 100% less the sum of sand and clay percentages. We model clay and sand, and set silt to 100% minus their sum, because clay and sand are the most important agronomically.

Because sand and clay values are imputed with separate regressions, their sum may occasionally add to more than 100%. In those cases, we scale each property relatively so that their sum adds to 100%, leaving silt in these fields at zero.

The sand and clay models are trained on an external dataset of point-level sand and clay measurements, synthetically constructed field-level composite sand and clay measurements, and point-level gSSURGO values for sand and clay.

The fitted linear regressions are:

$$\text{logit}(\texttt{clay}) = -2.923 + -0.017 \times \texttt{clay_ssurgo} + 0.080 \times \texttt{clay_composite} \tag{3a}$$

$$\text{logit}(\texttt{sand}) = -2.278 + 0.009 \times \texttt{sand_ssurgo} + 0.040 \times \texttt{sand_composite} \tag{3b}$$

Note that both models (3a) and (3b) put more "weight" on the composite sample taken in the field (0.80 and 0.40, respectively) than on the SSURGO prediction of texture at that point in the field. This greater weight confirms the value of taking ground-truth samples, even just one composite sample per field, rather than relying only on SSURGO. In Equation (3), the target variables—`clay` and `sand` content—are transformed with the logit function

$$\text{logit}(x) \equiv \log\left(\frac{x}{1-x}\right),$$

which is commonly applied to outcome variables that are constrained to be between zero and one (so that the predictions on the right-hand side can be any real number).

Table 18 shows that the coefficient of determination (R^2) value is large, so a lot of the variance is explained by this model. Because the same dataset is used to train both the clay and sand models, the model ID is identical.

Table 18: Summary of the soil pH imputation model.

Outcome variable:	clay	sand
R^2	0.557	0.796
Model ID (hash)	3182159020	3182159020
Fitted on	June 15, 2024	June 15, 2024

4.2.2 Predictive distribution

For each imputed sand and clay value, the predictive distribution is normally distributed that is centered at the mean sand (or clay) prediction of the linear regression model, with standard deviation equal to the square root of the calculated prediction variance for the predicted point. Table 19 summarizes that predictive distribution. The standard deviation of the predictive distribution is rather large compared to the typical values, indicating that the texture imputation model has considerable uncertainty. As for the other imputation models, this uncertainty is propagated to the biogeochemical model because we use a draw from the predictive distribution.

Table 19: **Summary of the predictive distribution** of the soil texture imputation. Abbreviation: SD = standard deviation.

Quantity	Outcome variable Summary statistic	clay	sand
mean prediction	average across points	24.1%	27.0%
	SD across points	11.6%	18.7%
	average of within-field SDs	5.7%	4.6%
SD of the predictive distribution	averaged across points	67.7%	61.8%
	SD across points	0.0%	0.0%
single draw from the predictive distribution	average across points	24.8%	27.5%
	SD across points	13.5%	19.3%
	average of within-field SDs	11.6%	8.9%

Like for imputations of pH (see Sec. 4.1.2), "extreme" values of texture drawn from the predictive distribution were rejected, and draws from the predictive distribution were repeated until a non-extreme value was drawn. These imputations of texture are used in Saxton's equations to compute inputs to the biogeochemical model (see supporting document `IndigoCarbon_US-1_2023_0030` (2023) for more details). Therefore, we impose the same constraints used by Saxton et al. (1986) on our imputations of texture: that $5\% < \text{clay} < 60\%$ and that $\text{sand} > 5\%$. These ranges cover most of the training dataset (described in Sec. 4.1.4): 5.2% of clay measurements were less than 5%; 3.4% of clay measurements were greater than 60%; and 0.1% of sand values were less than 5%.

4.2.3 Correlation with other soil properties

The texture imputation model does explicitly not capture correlations with soil properties other than texture, because the only predictors in Equation (3) are texture variables. Correlations among

sand, silt, and clay are somewhat captured by the constraint that sand, silt, and clay add to 100%.

4.2.4 Datasets used

For sand and clay imputations, we train our models with a static dataset of point-level sand and clay measurements using the same dataset used for pH (from Indigo Research Partners); please refer to Section 4.1.4 for a summary of that dataset. Because of the high costs of measuring texture in the lab (compared to pH), the Indigo Research Partners dataset has one texture measurement for every ten acres (i.e., one of out ten points in a one-acre grid sample design).

4.2.5 Number of points and fields at which soil texture is imputed

Because soil texture and pH are measured on the same composite samples, the number of points and fields at which we impute texture (Table 20) is identical to that for pH (Table 15).

Table 20: Summary of the number of soil texture measurements that were imputed. Abbreviation: SD = standard deviation.

Fields	6161
Carbon points per field (mean ± SD)	11.8 ± 8.1
Fields with composite samples	4590 (74.5%)

4.2.6 Variability of imputed and measured soil texture values

Table 21 shows the variability among imputed values in the Carbon by Indigo program (left-hand columns) and among the training dataset in Indigo Research Partners (right-hand columns). The inter-quartile range and within-field variability are similar. The imputation model's minima and maxima values are less extreme than the measured values, indicating that the model is skeptical of such extreme values.

Table 21: Summary of the variability of texture measurements and imputed values. Abbreviation: SD = standard deviation.

Texture measured in...	Carbon by Indigo		Indigo Partners	
Outcome variable	clay	sand	clay	sand
minimum	5.0%	5.0%	0.0%	2.0%
25th percentile	13.8%	13.4%	19.0%	22.0%
50th percentile	22.2%	21.0%	26.0%	31.0%
average	24.8%	27.5%	26.8%	37.2%
75th percentile	33.8%	35.7%	33.0%	49.0%
maximum	60.0%	92.1%	90.0%	97.0%
SD	13.5%	19.3%	14.2%	21.0%
average of the within-field SDs	11.6%	8.9%	6.7%	9.3%

4.2.7 Bias and accuracy in cross validation

Table 22: Summary of the performance of the soil texture imputation model on held-out data in cross validation. Shown is the average of the metric, plus or minus the standard deviation, across the $k = 10$ folds. Abbreviation: RMSE = root mean squared error.

Outcome variable:	clay	sand
RMSE	9.38% ± 0.11%	9.48% ± 0.16%
Bias	-1.72% ± 0.22%	-0.41% ± 0.24%

4.3 Methodology for predicting soil pH and texture

We use soil measurements from a separate Indigo business line, Indigo Research Partners, within a cross validation framework to test a suite of texture and soil pH prediction models. These models are evaluated for bias and performance before a final model for each soil property is selected. For use in the Carbon by Indigo, these final models are trained on all available data from Indigo Research Partners and used to make texture and pH predictions at all sample locations in the Carbon by Indigo program. In this section we walk through our analysis datasets, cross validation framework, considered models, and considered predictors. Results of this work are presented in the subsequent section.

4.3.1 Analysis dataset

For model selection, we use soil measurements from Indigo Research Partners, which provides us with point-level measurements for model fitting and testing. We are not able to use the data from the Carbon by Indigo program for model fitting because only composite, field-level measurements were taken.

We subset Indigo Research Partners data for pH and for texture separately. For each soil property, we first subset to all data points where that soil property was measured. We then further subset to fields with a minimum number of soil samples per field. That minimum was set for us to generate a synthetic composite measurement in each field at roughly the same sampling density in which actual composites were measured in the Carbon by Indigo program. Count of points used for model selection and model fitting are outlined in Table 23.

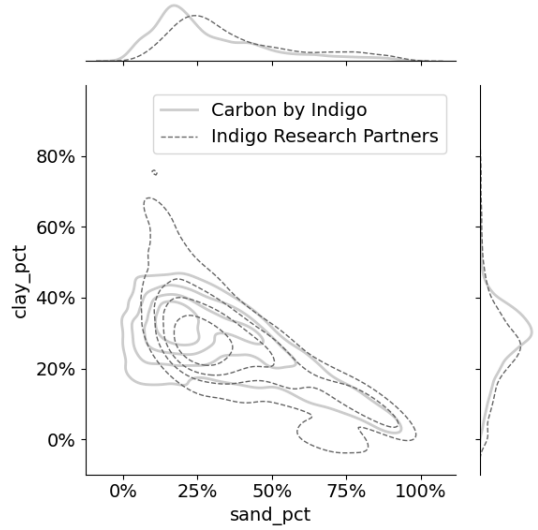
We note that there is an incomplete overlap in geographic coverage and in soil property values between the Carbon by Indigo dataset on which we predict soil properties and the Indigo Research Partners dataset with which we fit the model. In Figure 2 below we summarize the state overlap between these two soil sample datasets and also the values of those datasets

4.3.2 Cross validation framework

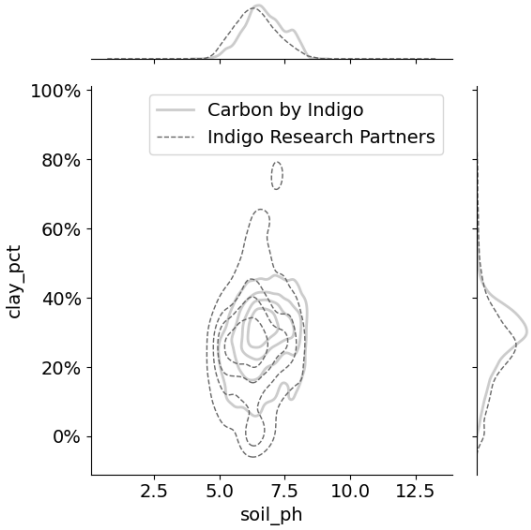
We use cross-validation experiments to select the model (among a menu of options described next in Sec. 4.3.3) and to evaluate how well it predicts out of sample. For our cross validation experiments we used sample-level measurements of pH and soil texture collected in the 2018 Indigo Research Partners sampling campaign. This allows us to simulate taking composite measurements and

Table 23: Filters applied to the Indigo Research Partners datasets for the pH model and for the texture model.

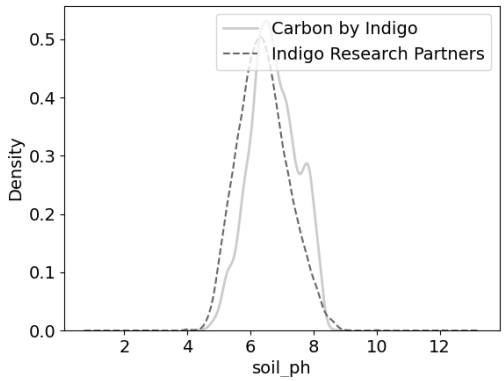
pH	Original data	67475	100.00%
	Limit to points with pH samples	47788	70.82%
	Limit to fields with > 20 pH samples	46310	96.91%
Texture	Original data	67475	100.00%
	Limit to texture samples	6157	9.12%
	Limit to fields with > 3 texture samples	6094	98.98%



(a) clay versus sand content



(b) clay content versus pH



(c) Histogram of pH

Figure 2: Comparison of Indigo Research Partners and the Carbon by Indigo program

validating our predictions at sample points. Our cross validation approach is structured in two steps. Each step is repeated 10 times:

Step 1: Cross validation for constructing a synthetic composite sample We first randomly select a fixed fraction of sample points from each field, p, from which to construct a synthetic composite sample. This composite sample is used as a predictor to train and test our regressions

Our objective is to select p that approximates the density of sample points taken in the Carbon by Indigo program for composite pH and texture measurements: between 1 sample / 24 acres and 1 sample / 40 acres (or one sample per every 3–5 carbon samples, where each carbon sample is taken every 8 acres).

We use the sampling rates in Indigo Research Partners to determine p for both Texture and pH. Table 24 highlights our selection of p.

Table 24: Fraction p of samples in Indigo Research Partners that were selected randomly to approximate the sample density in the Carbon by Indigo program for purposes of cross-validation experiments

Soil property	IRP median sampling rate	p used in cross validation
pH	1 sample per acre	$= 1/24 \approx 5\%$
Texture	1 sample per 10 acres	$= 10/24 \approx 40\%$

Step 2: Train/test splits We next randomly split each field into train and test datasets, selecting a fraction, q, to be in our training dataset. q is currently set to 50%. Figure 3 provides a visual overview of this selection schema.

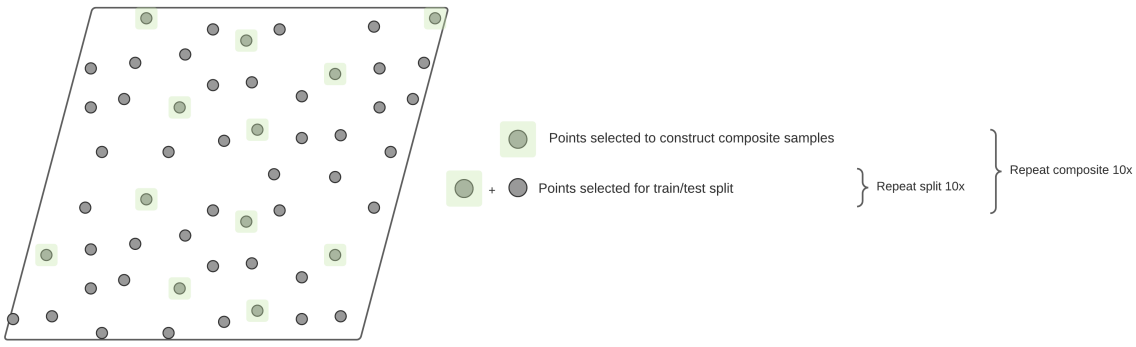


Figure 3: Illustration of the nested cross-validation used for pH and texture

4.3.3 Considered models

We compare all models considered for predicting pH and texture to each other and to the baseline scenarios of

1. using the composite sample for the field at each point and
2. using the gSSURGO estimate at each point.

The models we considered ranged from scaling gSSURGO, to linear regression models, logistic models, and simple machine learning algorithms. Table 25 summarizes the models we considered.

Table 25: Models considered for imputation of soil pH and texture

Model	About / why considered
Composite sample for whole field	Simplest interpolation approach serving as baseline
gSSURGO estimate at each sample point	Simple interpolation approach serving as baseline
Scale gSSURGO by ratio of composite sample to composite gSSURGO	Simple approach relying on available gSSURGO values at each point
Linear regression of gSSURGO points sample points	Straight-forward, easy to interpret relationship between gSSURGO and measured values. Does not rely on composite measurements
Linear regression of composite sample and point-level attributes	Straight-forward global estimator with composite values and point-level attributes
Logit-transformed linear regression of composite sample and point-level attributes	Transformed global estimator to account for the bounded nature of texture and pH values
Decision Tree regression of composite sample and point-level attributes	Simple machine learning algorithm for predicting soil properties

4.3.4 Considered predictors

In our evaluation of models to predict pH and texture, we considered several categories of predictors. We may also consider new predictors in future iterations of our bulk density predictions. Table 26 below describes these considered predictors.

4.3.5 Computing average pH

pH is calculated as the negative log of the concentration of hydronium ions (in moles per liter):

$$\text{pH} = -\log(h) \qquad \text{where } h = H_3O^+ \text{ concentration.} \tag{4}$$

Table 26: Predictors considered for imputation of soil pH and texture. Abbreviations: %SOC = soil organic carbon concentration; BD = bulk density; IRP = Indigo Research Partners

Predictor category	Predictor	About / why considered
Location	Latitude and longitude	Used for geographic trends and models that use proximal points for predictions
	x and y coordinates in the Albers projection	Same reasons as for latitude and longitude, but locations are transformed to an equal-area projection (in meters)
Soil measurements	Soil organic carbon	Measured %SOC at each IRP point
	Bulk density	Measured BD at each IRP point
gSSURGO	Bulk density Soil organic matter Texture Soil pH	Predictions are available across both Indigo Research Partners and the Carbon by Indigo program
Lidar 10m resolution from the NED (National Elevation Dataset)	Aspect Elevation Slope TPI (Topographic Position Index) TRI (Terrain Ruggedness Index)	Topography is an important property of soil that may be predictive of soil pH and texture
PTFs (Pedotransfer functions)	Pachepsky, Rawls, and Lin (2006) Abdelbaki (2018) Román Dobarco et al. (2019)	PTFs are used to predict certain soil properties from soil surveys

Therefore, our composite samples do not represent the average pH value at the sample locations. Instead, when n samples of equal volume are combined into a composite and then homogenized, the concentrations of hydronium ions in the samples are averaged, and our pH measurement is the negative log of that average concentration. Thus, we use the following transformation to simulate the pH measurement on a composite sample:

$$\overline{\text{pH}} = -\log\left(\overline{h}\right) = -\log\left(\frac{1}{n}\sum_{i=1}^{n} h_i\right) = -\log\left(\frac{1}{n}\sum_{i=1}^{n} 10^{-\text{pH}_i}\right), \tag{5}$$

where the sum runs across samples i that enter into the (simulated) composite.

4.4 Results of model selection for soil pH and texture

4.4.1 Selected approaches

For pH, we find that a linear regression model on composite pH samples and point-level pH gSSURGO values leads to a simple high-performing prediction of point-level pH. In our cross validation experiments, this approach produced the lowest root mean squared error of predicted values.

For texture, which is predicted with separate models for sand and clay, we find that a logit-transformed linear regression model on composite sand or clay samples (respectively) and point-level sand or clay gSSURGO values (respectively) lead to the highest performing prediction of point-level texture values.

While we identified a different model for each soil property to achieve the lowest RMSE, we opted to use similar models with similar predictor values for pH and texture. The linear regression and logit-transformed linear regression models with point-level gSSURGO and composite measurement predictors are easy to explain and all achieved low RMSEs relative to the broader field of considered models. Furthermore, these linear regressions have clear theoretical foundations for constructing predictive distributions. Table 27 summarizes the performance of our considered models

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Table 27: Results of cross validation experiments for predicting soil pH and texture. Performance of the selected models is highlighted in green.

	pH		sand		clay	
	Bias	RMSE	bias	RMSE	bias	RMSE
Composite sample for whole field	-0.267	0.718	-0.026	9.594	-0.041	7.272
gSSURGO estimate at each sample point	-0.207	0.887	-12.274	19.362	0.555	12.336
Scale gSSURGO by ratio of composite sample to composite gSSURGO	-0.201	0.741	0.751	22.469	0.093	8.374
Linear regression of gSSURGO points	-0.000	0.758	0.107	13.796	-0.218	11.182
Linear regression of composite sample and gSSURGO points	-0.001	0.641	0.004	9.407	-0.008	7.064
Linear regression of Lasso-optimal predictors	-0.002	0.636	0.017	9.417	0.020	7.069
Logit-transformed linear regression of composite samples and gSSURGO points	N/A	N/A	-0.408	9.491	-1.575	9.490
Decision Tree regression of composite sample + gSSURGO points	-0.001	0.563	-0.026	10.495	0.001	7.405

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Acronyms

IRP A program involving field trials and other research at Indigo Ag. 15, 16, 19–22, 24

NED A national elevation dataset made by the USGS and other partners 24

PTF Pedotransfer functions are predictive functions of certain soil properties using data from soil surveys. Most PTFs are developed to predict soil hydraulic properties such as hydraulic conductivity or water retention. 24

TPI Topographic Position Index compares the elevation of each cell in a DEM to the mean elevation of a specified neighborhood around that cell. 24

TRI The mean of the absolute differences between the value of a slope in a cell and the value of its 8 surrounding cells 24